Math 2650 – Quiz 06 Concept Practice

Choose the section matching the skill you want to practice. Answers are on the back.

A. Taylor Polynomials and Linear Approximations

(a) Suppose $f(x, y)$ has second-order Taylor polynomial

$$P_2(x, y) = -2 + 4x + y - 5x^2 + 6xy + 3y^2$$

at $(0, 0)$. Find $f_x(0, 0)$, $f_{xx}(0, 0)$, and $f_{xy}(0, 0)$.

(b) Find the cubic Taylor approximation of

$$f(x, y) = e^{2x} \cos y$$

at $(0, 0)$.

(c) Suppose $g(x, y)$ has second-order Taylor polynomial

$$P_2(x, y) = 3 + 2x - y + x^2 + 2xy - 3y^2$$

at $(1, 2)$. Find the linear approximation to g at $(1, 2)$.

B. Normal Lines and Tangent Planes

(a) Find parametric equations for the normal line to

$$x^2y + e^{xz} + yz^2 = 2$$

at the point $(1, 1, 0)$.

(b) Consider the surface

$$x^3 + y^2 - 3x + 2y - 2z = 1.$$

Find all points P on the surface for which $\mathbf{v} = \langle 9, -4, -2 \rangle$ is normal to the tangent plane at P.

C. Critical Points and Classification

(a) Find and classify the critical point of

$$f(x, y) = \ln(x + 2y) + x^2 - 2y.$$

(b) Find a and b so that

$$f(x, y) = ax^3 + xy + by^4$$

has a critical point at $(1, 1)$, then classify that critical point.

(c) Find and classify all critical points of

$$h(x, y) = x^4 - x^2 + 2xy + y^2.$$

D. Linearization and Directional Estimates

(a) The volume of an ellipsoid is

$$V(a, b, c) = \frac{4\pi}{3}abc.$$

Use a linearization at $(2, 3, 4)$ to estimate the volume when

$$(a, b, c) = \left(\frac{21}{10}, \frac{57}{20}, \frac{17}{4}\right).$$

(b) Use a directional derivative to approximate the change in

$$f(x, y, z) = xe^z + y \cos z + x - y$$

when moving from $P_0 = (1, 2, 0)$ a distance $ds = 0.4$ toward $P_1 = (3, 1, 2)$.

Answer Key

Final answers only

A. Taylor Polynomials and Linear Approximations

(a)

$$\boxed{f_x(0,0) = 4, \qquad f_{xx}(0,0) = -10, \qquad f_{xy}(0,0) = 6}.$$

(b)

$$\boxed{P_3(x,y) = 1 + 2x + 2x^2 - \frac{1}{2}y^2 + \frac{4}{3}x^3 - xy^2}.$$

(c)

$$\boxed{L(x,y) = -4 + 8(x-1) - 11(y-2)}$$

or equivalently

$$\boxed{L(x,y) = 8x - 11y + 10}.$$

B. Normal Lines and Tangent Planes

(a)

$$\boxed{x = 1 + 2t, \qquad y = 1 + t, \qquad z = t}.$$

(b)

$$\boxed{P = (2,-3,2) \quad \text{or} \quad P = (-2,-3,0)}.$$

C. Critical Points and Classification

(a)

$$\boxed{\left(-\frac{1}{2}, \frac{3}{4}\right) \text{ is a saddle point}}.$$

(b)

$$\boxed{a = -\frac{1}{3}, \qquad b = -\frac{1}{4}}$$

and

$$\boxed{(1,1) \text{ is a local maximum}}.$$

(c)

$$\boxed{(0,0) \text{ is a saddle point}}$$

and

$$\boxed{(1,-1) \text{ and } (-1,1) \text{ are local minima}}.$$

D. Linearization and Directional Estimates

(a)

$$\boxed{V \approx 34\pi}.$$

(b)

$$\boxed{\Delta f \approx \frac{4}{5}}.$$